

Climate vulnerability of the world's amphibians

Finding the missing pieces of an ecological puzzle



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SEB



BACKGROUND

- Extreme heat events are predicted to increase in both **frequency and intensity**
- Thermal tolerance** is an important trait to assess the vulnerability of animals to climate change
- However, **we know very little** about the thermal tolerance of **most amphibians species**
- AIM 1:** Compile the **most comprehensive database** on amphibian thermal tolerance
- AIM 2:** Use this database to **infer the thermal tolerance and climate vulnerability of most amphibians species** using data imputation (*in progress*)



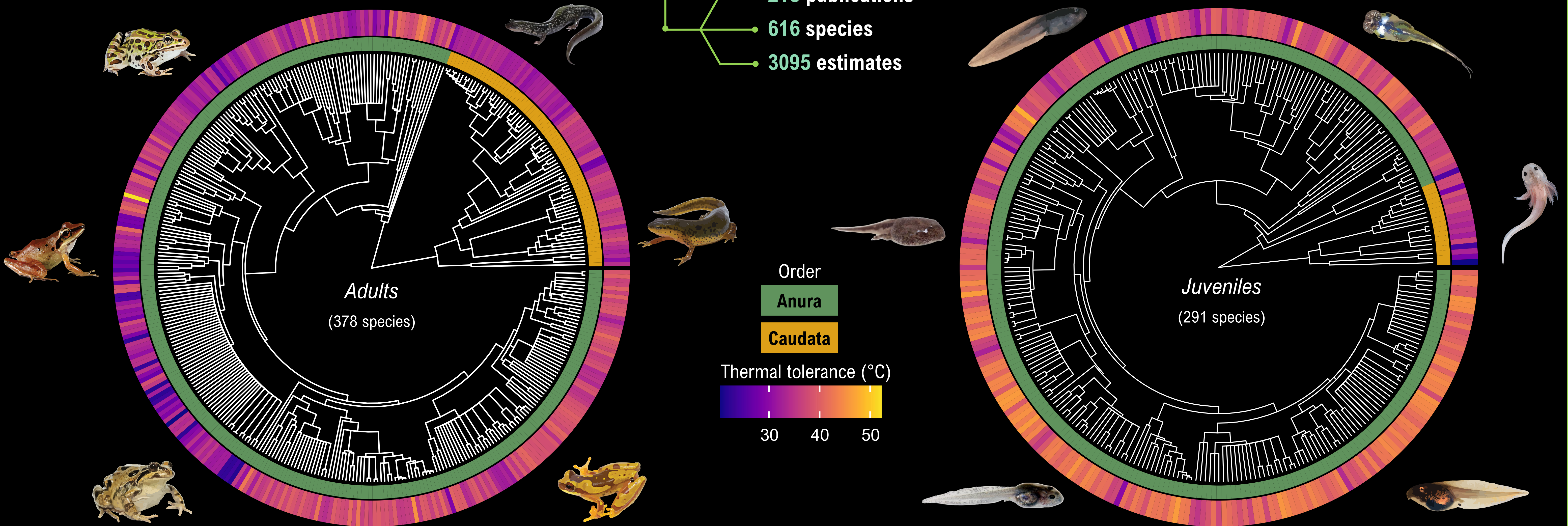
SYSTEMATIC REVIEW

- Criteria
 - Juvenile or adult amphibian
 - CTmax or LT50 as proxies for thermal tolerance
 - Temperature of acclimation reported in the study
- 5 databases scrapped in 7 languages (English, Spanish, Portuguese, Japanese, French, simplified Chinese, traditional Chinese)
- >4000 publications screened



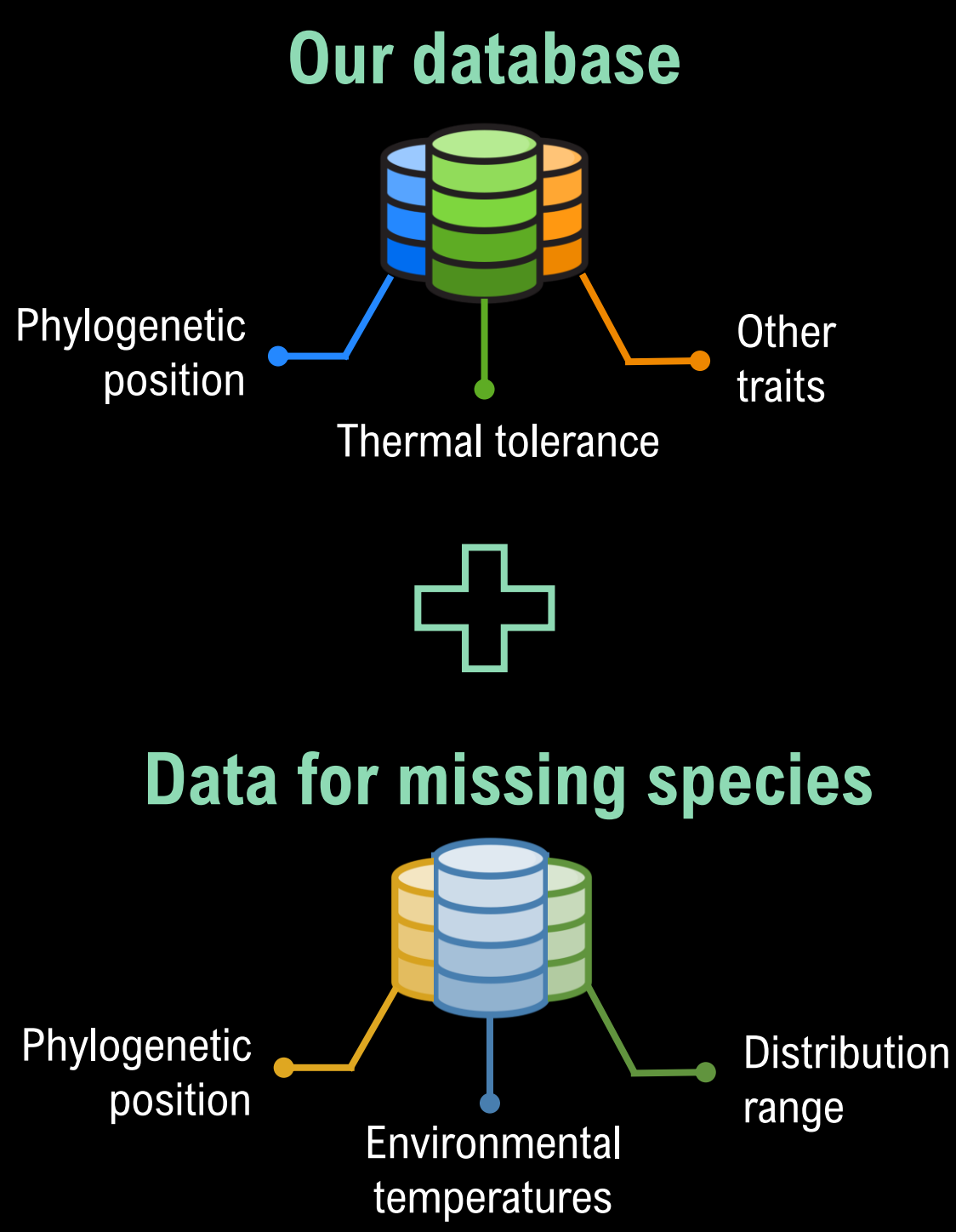
THE DATABASE

- 213 publications
- 616 species
- 3095 estimates



NEXT STEPS

- Although our data synthesis is comprehensive, we could only retrieve data for **616 species out of the ~8500 currently described**
- We will use powerful imputation procedures to **retrieve the thermal tolerance of most missing species** and **infer their vulnerability to climate extremes**



Multiple data imputation



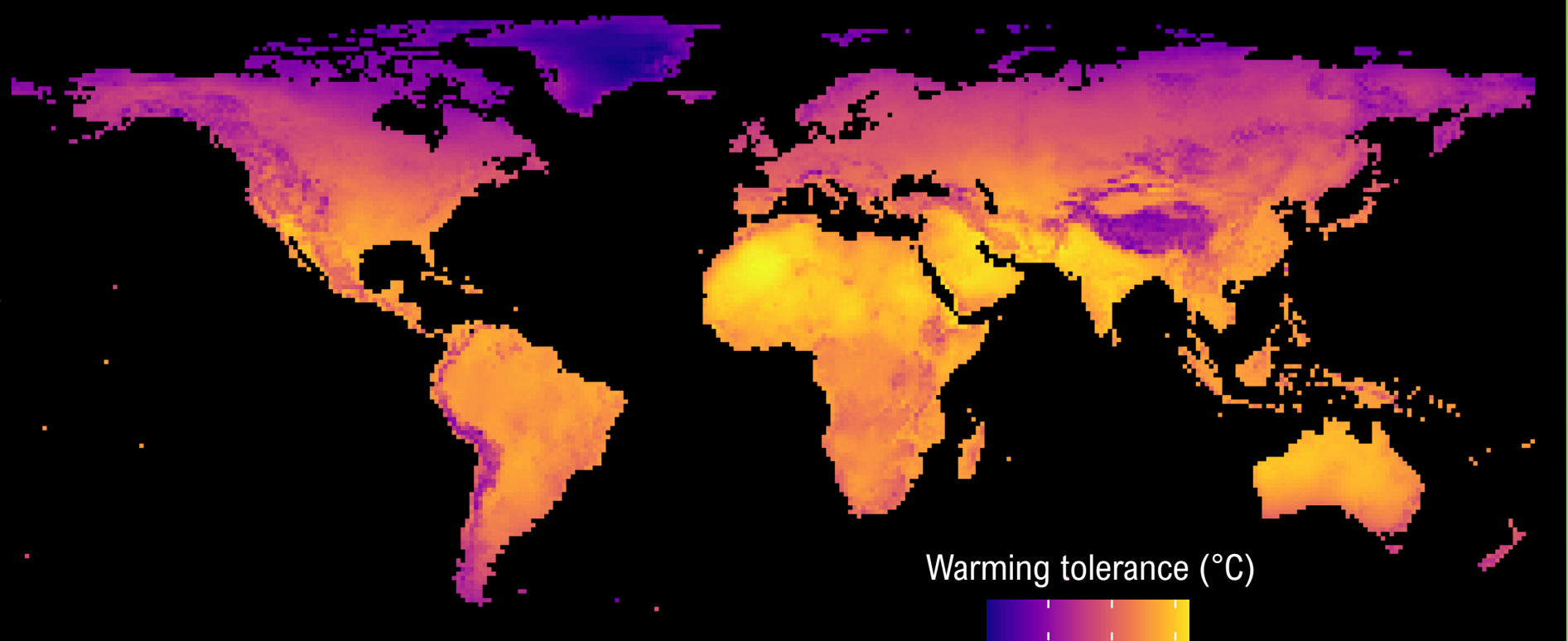
hmi package

Infer the **thermal tolerance of missing species**

Inferred thermal tolerance

Environmental temperatures

Estimate the **climate vulnerability of most amphibian species**



SIGNIFICANCE

We compiled the **most comprehensive database on amphibian thermal tolerance**, which can be use to address various questions in ecophysiology

Our next step will infer the thermal tolerance of **thousands of missing species, without the need to collect new data**

Transposing thermal tolerance against experienced environmental temperatures will **identify the locations and species most vulnerable to rising temperatures**

Our study will provide essential evidence for **the conservation of amphibians in the Anthropocene**

Acknowledgements



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Bedegal people



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